Andrew Klesh

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RESEARCH INTERESTS

- In-situ exploration of extreme environments (deep space, arctic, volcanic, glacial, underwater)
- Low-cost and NanoSpacecraft mission development and operations

EXPERIENCE

JPL (2011-Present)

Leadership in Project and Mission Development & Execution

- **Brinson Exploration Hub** – Associate Director / Chief Systems Engineer (2024-Present) for joint Caltech/JPL "skunkworks", rapidly executing high-risk/high-reward expeditions and missions.
- **Lunar Trailblazer** – Proj Sys Engineer (Lead) (2020-2025) for ˜$86M 210kg lunar s/c with JPL & international instruments & multi-institution integration & ops.
- **Ocean Exploration / Buoyant Rover for Under-Ice Exploration (BRUIE)** – Engineering lead of novel systems to explore ice-water interface & Hadal deep ocean with WHOI. More than 30 arctic, Antarctic, ocean ice-breaker, and mountaineering scientific expeditions.
- **Distributed Acoustic Sensing (DAS)** – Collaboration w/ Caltech for field deployment of fiber-optic seismic networks including Iceland; Kilauea, HI; Taku Glacier, AK; & Ridgecrest, CA. Geology master's thesis: "Improving Knowledge of Lava Pathways at Kilauea Using DAS".
- **MarCO** – Chief Engineer (2014-2020). Responsible for technical development, I&T, and ops of the first interplanetary CubeSats, leading to successful 2018 InSight EDL support at Mars.
- **INSPIRE** – PI and Chief Engineer (2012-2014). Led development of the first deep space CubeSats & technology (including Iris radio, Nano Thin Star Tracker & protos FSW).
- **Mission Architect / Systems Engineer** – Have consulted for, developed, or led ¿20 robotic & small spacecraft missions throughout mission lifecycle and supported numerous special studies for large projects, including Europa Lander immolation system.

EDUCATION

- **M.S. Geology** (CalPoly-Pomona – 2023)
- **Postdoc:** JAXA Hayabusa / Ikaros (2010)

U. Michigan (2001-2010)

- **Postdoc:**RAX CubeSat (2009)
- **Ph.D. Aerospace Engineering** –FDC (2009)
- **M.Eng Space Systems Engineering** (2009)
- **M.S.E Aerospace Engineering** (2006)
- **B.S.E. Aerospace Engineering**, cum laude (2005)
- **B.S.E Electrical Engineering**, cum laude (2005)

TEACHING & MENTORING

JPL, Caltech and Arizona State University:

- Macgyver Engineering (Field Engineering)
- EE154: Practical Electronics for Space
- Deep Space Ops for Small Spacecraft (Remote Multi-Institutional Course)

SELECT PROFESSIONAL DISTINCTIONS

- 2020 AIAA Engineer of the Year
- 2019 NASA Exceptional Engineering Medal
- 2016 JPL Lew Allen Award for Excellence
- 2010 IAF Luigi G. Napolitano Award

ADDITIONAL RELEVANT SKILLS & ONGOING ACTIVITIES

- Private Pilot (IFR) with Tailplane & Skiplane Training. UAS Operator Licensed.
- 2 x National Outdoor Leadership School – Alaska Mountaineering / Glacier Rescue + Denali
- Scuba Diver (Rescue) (350+ dives incl. Dry-Suit, Ice & Altitude) + Hyperbaric Chamber Supervisor
- Pasadena Fire Department EMSR EMT / AEMT (+ Wilderness EMT & JPL USAR Capt.) (¿4400 hrs)